
AMBIENT
SYSTEMS
Reimagining the future of smart cities


40%
of **global CO2 emissions** is attributed to buildings

THE PROBLEM

Climate impact will largely be felt by building owners and occupants



$260B+

of portfolio losses due to climate change

climate risk & operational inefficiency can reduce asset value by 20% and annual returns by 40%

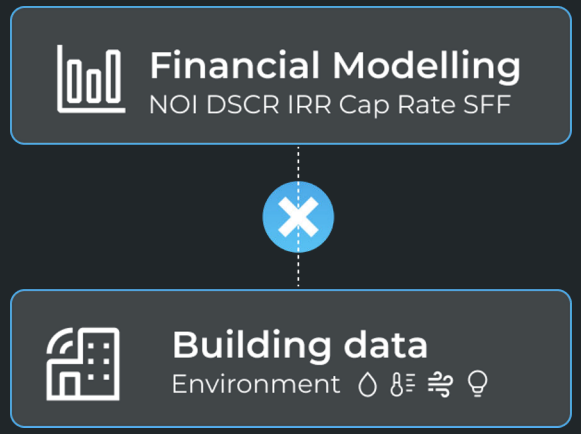
CURRENT SOLUTIONS

Decarbonization solutions for Commercial Real Estate are missing key things



01

Climate Targets not Fully Included



02

Financial Modelling disconnected from Building Data

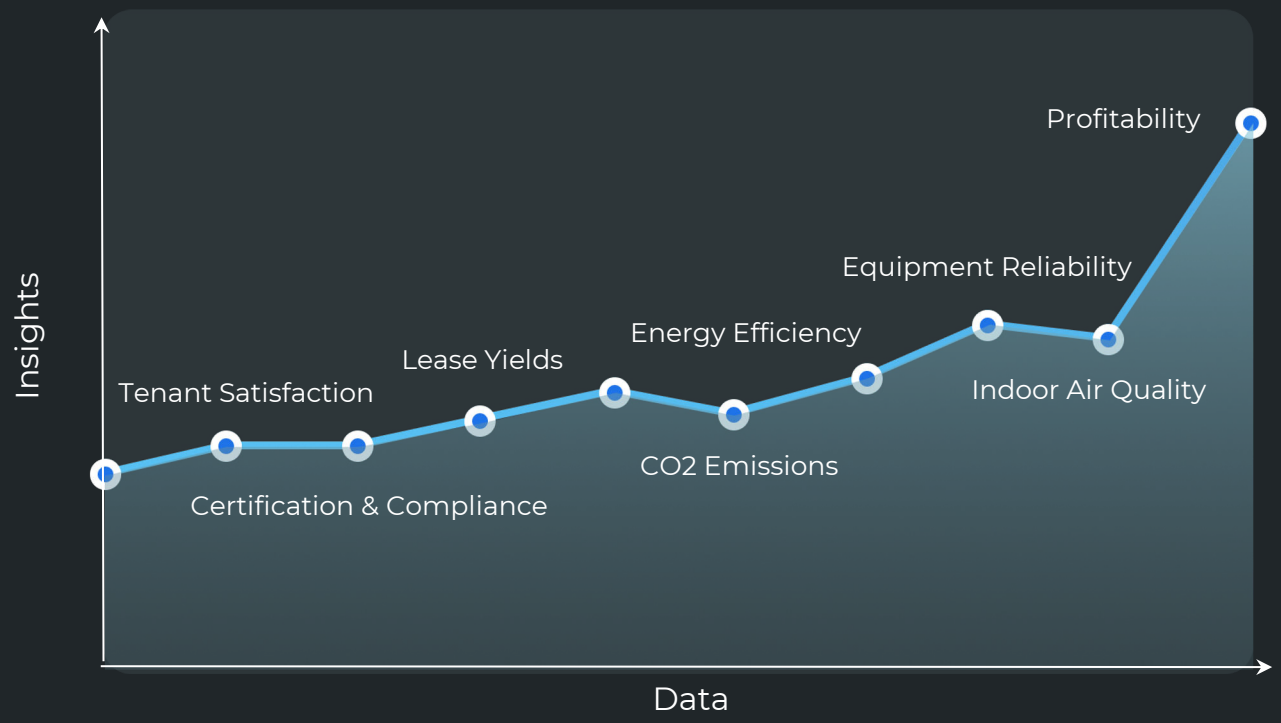


03

Hard to Find Analytical Expertise

AMBIENT'S SOLUTION

Driving real estate decarbonization through data science



Secure green capital and tenants by:

01 Streamlining data collection & analysis

02 Quantifying embodied & operational emissions

03 Identifying decarbonization pathways

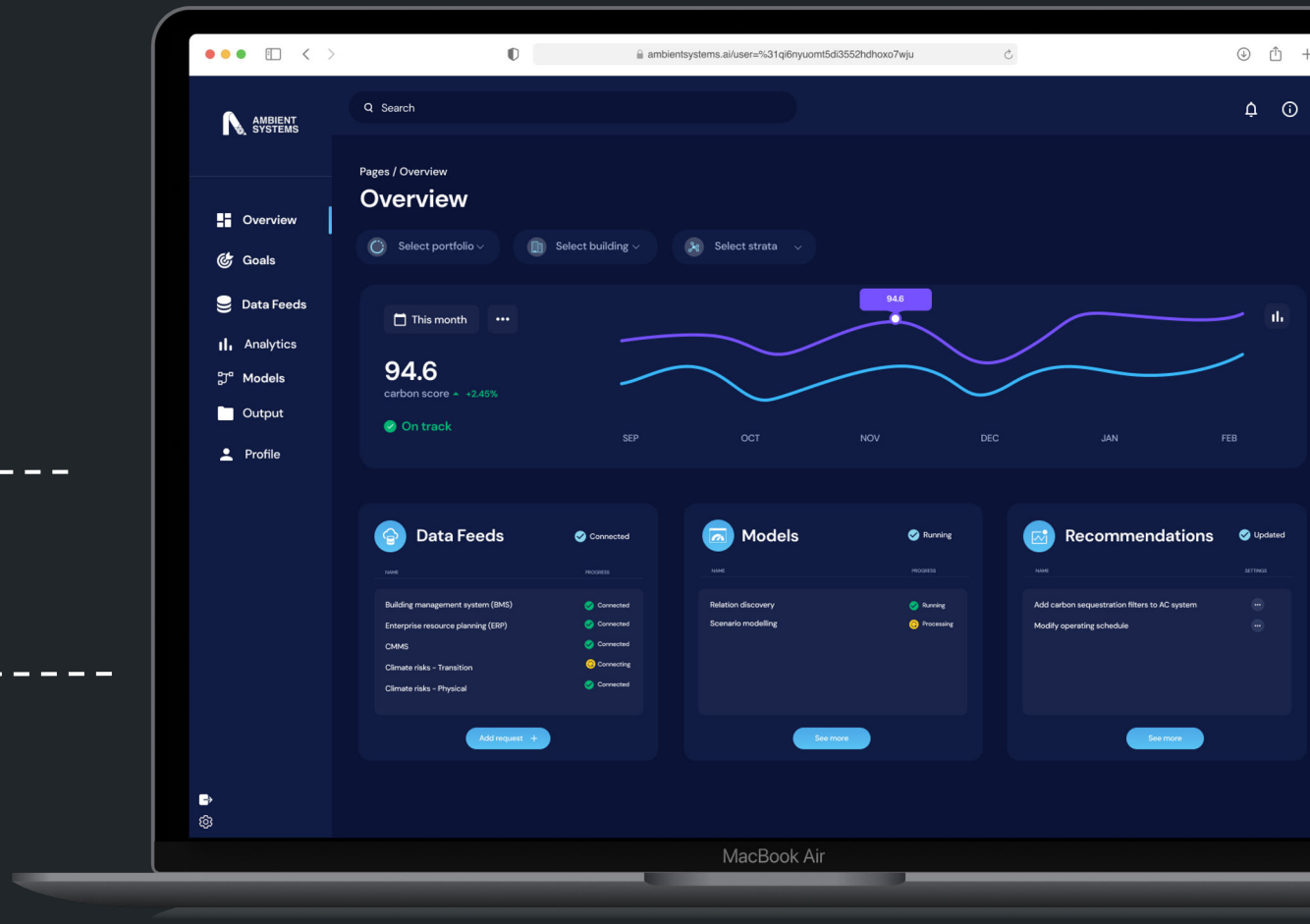
PRODUCT

All-in-one Decarbonization Platform for Real Estate – powered by the Cloud & Data Science

AUTOMATE LIFE CYCLE ASSESSMENT
TO QUANTIFY CARBON FOOTPRINT

IDENTIFY NET ZERO PATHWAYS
ACROSS DESIGN & OPERATIONS

DELIVER INSIGHTS & REPORTS TO
STAKEHOLDERS



UNIQUE SELLING POINT

Ambient provides a tailored SaaS offering for sustainable real estate



Flexible On-demand Subscription

No more onboarding, training and manpower reliance. Fast up-time and results through Ambient's subscription service



Simple and Affordable Pricing Plans

No more reliance on expensive consultants or in-house teams. One platform powered for all your data science use cases



Specialized AI Models built on Domain Expertise

Reach business goals across energy efficiency, environment, and compliance with Ambient's state of the art AI models

HOW IT WORKS

Ambient delivers net zero pathways directly to front-line service teams

Data

Technical

- BIM
- BMS
- CMMS
- BOM

Risk

- Energy & Commodity Prices
- Climate Risks (Direct & Transition)
- Labour & Materials Costs

Operational

- Electricity
- Water
- Gas
- Fuel

Data Science

Data Engineering Module

- BIM Processing
- BMS Integration
- CMMS Classification

- Graph Mining
- Video Analytics
- Prediction & Optimization

Output Delivery Module

- Controls automation
- Notifications & Emails

Pathways

Existing Buildings
Operational Carbon
(51% of emissions)

- Demand Shaping
- Electrification
- Efficiency
- DERs Deployment

New Buildings
Embodied Carbon
(49% of emissions)

- Passive Design
- Material Alternatives
- Efficient Design
- Construction Electrification
- Materials Routing
- Design for Recycling

Users

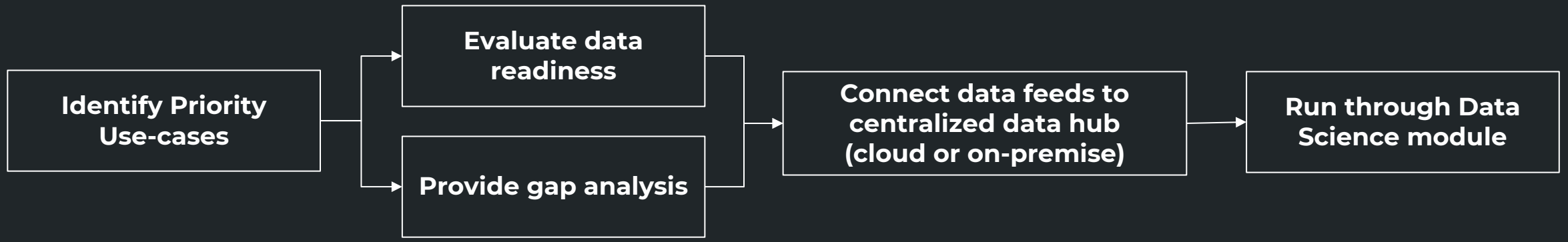
FM Teams
(Universities, Hospitals, Malls)

Property Management Firms
(REITs and Asset Managers)

Disclosure & Reporting Frameworks (Optional)

IMPLEMENTATION

Ambient identifies priority use cases then works backwards to identify needed datasets



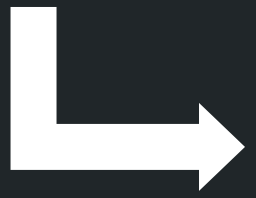
1. Decarbonization (real estate properties)
2. Decarbonization (transit)
3. Retail Analytics
4. Revenue Optimization

1. Data Engineering
2. [redacted]
3. Output delivery (controls automation, notification, email, reports, or external data feed API)

HOW DOES IT WORK

[redacted] provides recommendations based on operational data

Date/Time	Electricity:Facility [kW](Hourly)	Fans:Electricity [kW](Hourly)	Cooling:Electricity [kW](Hourly)	Heating:Electricity [kW](Hourly)	InteriorLights:Electricity [kW](Hourly)	InteriorEquipment:Electricity [kW](Hourly)	Gas:Facility [kW](Hourly)	Heating:Gas [kW](Hourly)	InteriorEquipment:Gas [kW](Hourly)	Water Heater:WaterSystems:Gas [kW](Hourly)
01/01 01:00:00	174.2145729	32.75032584	48.42562787	0	34.43116061	33.10218689	154.7941419	28.41378994	3.33212	123.0482319
01/01 02:00:00	168.3524558	32.75032584	42.55659868	0	34.43116061	33.10218689	116.5007823	24.8622231	3.33212	88.30643925
01/01 03:00:00	146.3612177	32.75032584	38.07911969	0	19.46627941	30.61066878	119.1753379	22.2889279	3.33212	93.55428996
01/01 04:00:00	144.0482316	32.75032584	35.77850208	0	19.46627941	30.61066878	117.909866	20.51268543	3.33212	94.06506054
01/01 05:00:00	151.1611995	32.75032584	36.3191083	0	19.46627941	30.61066878	145.3588048	19.15022631	3.33212	122.8764585
01/01 06:00:00	165.1205174	32.75032584	38.04378529	0	19.46627941	30.61066878	180.6108081	18.307706	8.3303	153.9728021
01/01 07:00:00	[redacted]	35.03194464	45.96961404	0	42.69180509	54.3732441	[redacted]	17.7994769	16.6606	[redacted]
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01/01 10:00:00	[redacted]	37.88787676	102.6893755	0	50.95244957	81.16842718	[redacted]	42.92042145	66.8089	[redacted]



Demand Shaping | Electrification | Efficiency | DERs Deployment

COMPETITIVE ANALYSIS

Ambient provides the most intelligent decarbonization platform for real estate



DIFFERENTIATION

Ambient aims to be a leader in service, cost, and customization

	Ambient	Consultancy	In-house
Service	Ongoing Subscription with Dedicated Account Engineer	Fixed Project Duration Limited Maintenance Service Multiple Points of Contact	Hard to recruit Requires on-boarding Expensive training Management overhead High turnover risk
Cost	Fixed	Variable	Variable
Modelling	Based on domain expertise	Limited domain expertise	Requires multiple hard-to-find experts to tune Technical uncertainty and risk Expensive and complex IT vendor management process Cybersecurity risks

COMPETITIVE ADVANTAGE

Deep domain expertise in energy and engineering with connections to a global network

Domain Expertise

1. Saved $130MM of annual energy costs for world's biggest gold mining company using Big Data
2. Building Science know-how for efficient envelope design
3. Developed 200+ solar energy projects in Asia

Engineering Tech Stack

1. Built using leading cloud vendors (AWS, GCP, and Azure)
2. Embedded machine learning (Jupyter, MATLAB, Tableau)
3. Partnership with synthetic data companies to augment existing datasets

Global Partner Network

1. Member of United Nations & One Young World Ambassador Network for Sustainable Energy
2. Leveraging climate risk data from Switzerland's largest insurer
3. ASHRAE and IFMA Member

MARKET POTENTIAL

Opportunity for decarbonization of commercial real estate



$2.6B

Serviceable Obtainable Market (SOM)
Ambient's Potential Market Share

$26B

Serviceable Available Market (SAM)
Customers ready for adoption within next 3-5 years

$260B

Total Addressable Market (TAM)
Potential global losses due to climate change transition

APPLICATIONS

Ambient supports a broad range of CRE assets



Industrial

Manufacturing
R&D
Life Sciences and BioTech



Retail

Malls
Shopping Centers
Grocery Stores



Offices

High Rise
Low / Mid Rise
Data Centers



Transportation & Logistics

Public Transit
Airports
Warehouses and DCs

ABOUT US

The team behind Ambient

CEO
Deep tech entrepreneur with 3+ years experience in clean tech and machine learning

UX/UI Designer
Service & UIUX Designer with background in VC & Industrial Design

Data Scientist
MIT Machine learning engineer with experience building cross-industry AI applications at Meta and AWS

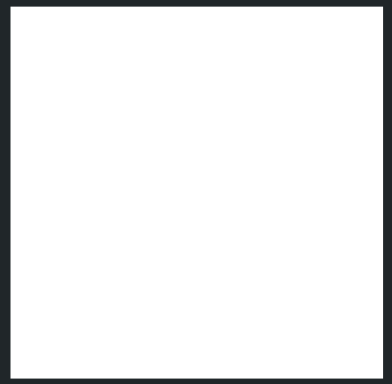
Developer
Experienced developer of gaming apps and web applications

Mentors & Advisors

Dr.
Experienced technology executive, investor, and educator with 35 years of leadership experience at Oracle and Hewlett-Packard

TRACTION

Backed by leading local institutions and intellectual property



Ambient Systems

- Launched in June 2021 in Singapore
- Deep tech startup focused on Machine Learning
- Founder sole inventor of Singapore patent (10202105064T)
- International team with members in Singapore, Toronto, Boston, and Belgrade
- $100k funding from National University of Singapore investment arm
- Agreements with 2 companies in Singapore + 1 in negotiation
- Domain expertise in civil engineering, construction, and cyber physical systems
- Team background in computer science, engineering, finance, and clean tech



VISION

Ambient empowers real estate leaders to achieve net zero portfolio growth through data science innovation

